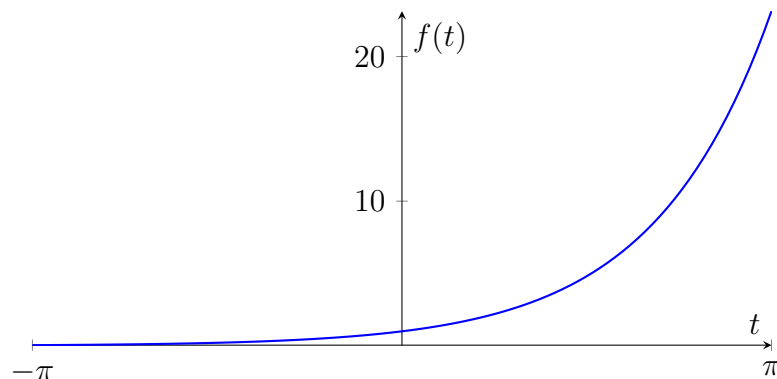


ECE 3210 hw04

1. Find the complex Fourier series of the 2π -periodic signal defined by $f(t) = e^t$ for $-\pi \leq t < \pi$.



- (a) Find the fundamental frequency ω_0 .

Solution: Since $T_0 = 2\pi$,

$$\omega_0 = \frac{2\pi}{T_0} = 1.$$

- (b) Find the complex Fourier coefficients D_n in simplest form.

Solution:

$$\begin{aligned} D_n &= \frac{1}{2\pi} \int_{-\pi}^{\pi} e^t e^{-jnt} dt \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{(1-jn)t} dt \\ &= \frac{1}{2\pi} \left. \frac{e^{(1-jn)t}}{1-jn} \right|_{-\pi}^{\pi} \\ &= \frac{1}{2\pi} \left[\frac{e^{(1-jn)\pi}}{1-jn} - \frac{e^{-(1-jn)\pi}}{1-jn} \right] \\ &= \frac{1}{2\pi(1-jn)} (e^{\pi} e^{-jn\pi} - e^{-\pi} e^{jn\pi}) \\ &= \frac{(-1)^n}{2\pi(1-jn)} (e^{\pi} - e^{-\pi}) \\ &= \frac{(-1)^n}{1-jn} \frac{\sinh(\pi)}{\pi}. \end{aligned}$$

- (c) Represent the Fourier series in terms of D_n .

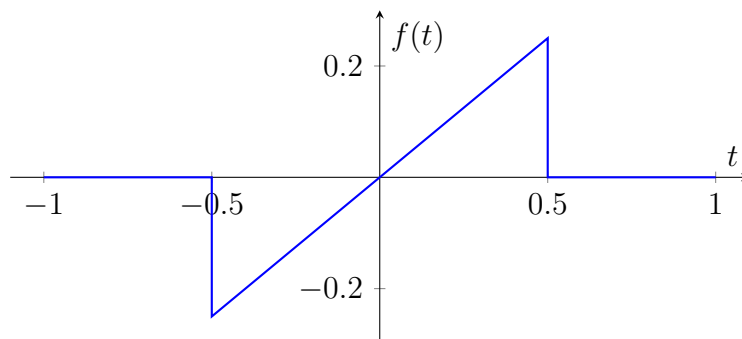
Solution:

$$f(t) = \sum_{n=-\infty}^{\infty} \frac{(-1)^n \sinh(\pi)}{\pi(1 - jn)} e^{jnt}.$$

2. A 2-periodic signal has one period described by

$$f(t) = \begin{cases} t/2, & -1/2 \leq t < 1/2, \\ 0, & 1/2 \leq |t| < 1. \end{cases}$$

Find its complex Fourier series in simplest form.



- (a) Find ω_0 .

Solution: Since $T_0 = 2$,

$$\omega_0 = \frac{2\pi}{T_0} = \pi.$$

- (b) Find D_0 and D_n for $n \neq 0$.

Solution: For $n \neq 0$,

$$\begin{aligned} D_n &= \frac{1}{2} \int_{-0.5}^{0.5} \frac{t}{2} e^{-j\pi n t} dt \\ &= \frac{1}{4} \left[\frac{t e^{-j\pi n t}}{-j\pi n} + \frac{e^{-j\pi n t}}{\pi^2 n^2} \right]_{-0.5}^{0.5} \\ &= \frac{1}{4} \left[\frac{0.5 e^{-j\pi n/2}}{-j\pi n} + \frac{e^{-j\pi n/2}}{\pi^2 n^2} - \frac{-0.5 e^{j\pi n/2}}{-j\pi n} - \frac{e^{j\pi n/2}}{\pi^2 n^2} \right] \\ &= \frac{1}{4} \left[\frac{0.5 e^{-j\pi n/2} + 0.5 e^{j\pi n/2}}{-j\pi n} + \frac{e^{-j\pi n/2} - e^{j\pi n/2}}{\pi^2 n^2} \right] \\ &= \frac{j}{4\pi n} \left(\cos\left(\frac{\pi n}{2}\right) - \frac{2}{\pi n} \sin\left(\frac{\pi n}{2}\right) \right). \end{aligned}$$

For the zero-frequency coefficient,

$$D_0 = \frac{1}{2} \int_{-1/2}^{1/2} \frac{t}{2} dt = 0.$$

- (c) Represent the Fourier series $f(t)$ in terms of D_n .

Solution:

$$f(t) = \sum_{n \neq 0} \frac{j}{4\pi n} \left[\cos\left(\frac{\pi n}{2}\right) - \frac{2}{\pi n} \sin\left(\frac{\pi n}{2}\right) \right] e^{jn\pi t}.$$

3. Consider

$$f(t) = 0.5 + 5 \sin(0.75t) + 2 \cos(t) + 7 \cos(-0.5t) + \sin(-0.25t).$$

Write the trigonometric Fourier series using fundamental frequency ω_0 and determine the requested coefficients.

(a) Find the fundamental frequency ω_0 .

Solution: The smallest angular frequency present is $\omega_0 = 0.25$.

(b) Find a_0 .

Solution: The constant term is $a_0 = 0.5$.

(c) Find $(a_1, a_2, a_3, a_4, a_5)$.

Solution: The cosine terms are $7 \cos(2\omega_0 t)$ and $2 \cos(4\omega_0 t)$, so

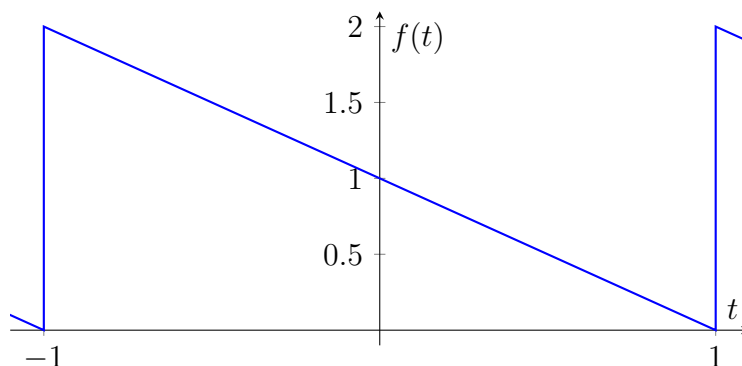
$$(a_1, a_2, a_3, a_4, a_5) = (0, 7, 0, 2, 0).$$

(d) Find $(b_1, b_2, b_3, b_4, b_5)$.

Solution: Using $\sin(-x) = -\sin(x)$, the sine terms give

$$(b_1, b_2, b_3, b_4, b_5) = (-1, 0, 5, 0, 0).$$

4. Find the complex Fourier series of the 2-periodic sawtooth signal defined on $(-1, 1)$ by $f(t) = 1 - t$.



- (a) Find ω_0 .

Solution: Since $T_0 = 2$,

$$\omega_0 = \frac{2\pi}{T_0} = \pi.$$

- (b) Find D_0 and D_n for $n \neq 0$.

Solution: Let $\tilde{f}(t) = 1 - f(t) = t$. For $n \neq 0$,

$$\begin{aligned} \tilde{D}_n &= \frac{1}{2} \int_{-1}^1 t e^{-j\pi n t} dt \\ &= \frac{1}{2} \left[\frac{t e^{-j\pi n t}}{-j\pi n} + \frac{e^{-j\pi n t}}{\pi^2 n^2} \right]_{-1}^1 \\ &= \frac{1}{2} \left[\frac{e^{-j\pi n}}{-j\pi n} + \frac{e^{-j\pi n}}{\pi^2 n^2} - \left(\frac{-e^{j\pi n}}{-j\pi n} + \frac{e^{j\pi n}}{\pi^2 n^2} \right) \right] \\ &= \frac{(-1)^n}{j\pi n}. \end{aligned}$$

Using $f(t) = 1 - \tilde{f}(t)$,

$$\begin{aligned} f(t) &= 1 - \sum_{n \neq 0} \tilde{D}_n e^{j\pi n t} \\ &= 1 + \sum_{n \neq 0} \frac{(-1)^n}{j\pi n} e^{j\pi n t}. \end{aligned}$$

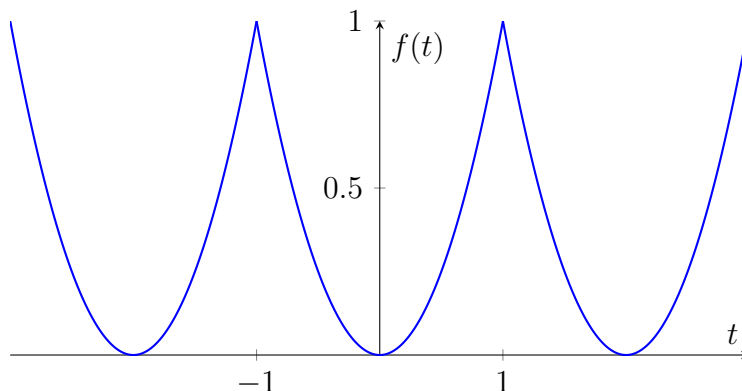
The average value is $D_0 = 1$.

- (c) Represent $f(t)$ as a complex Fourier series.

Solution:

$$f(t) = 1 + \sum_{n \neq 0} \frac{(-1)^n}{j\pi n} e^{jn\pi t}.$$

5. Let $f(t)$ be periodic with period 2, and let $f(t) = t^2$ on $-1 \leq t \leq 1$.



- (a) Find the period and fundamental frequency.

Solution: The period and fundamental frequency are

$$T_0 = 2, \quad \omega_0 = \frac{2\pi}{T_0} = \pi.$$

- (b) Find the complex Fourier coefficients.

Solution: For $n \neq 0$,

$$\begin{aligned} D_n &= \frac{1}{2} \int_{-1}^1 t^2 e^{-j\pi n t} dt \\ &= \frac{1}{2} \left[\frac{t^2 e^{-j\pi n t}}{-j\pi n} + \frac{2t e^{-j\pi n t}}{\pi^2 n^2} + \frac{2e^{-j\pi n t}}{j\pi^3 n^3} \right]_{-1}^1 \\ &= \frac{1}{2} \left[\frac{e^{-j\pi n}}{-j\pi n} + \frac{2e^{-j\pi n}}{\pi^2 n^2} + \frac{2e^{-j\pi n}}{j\pi^3 n^3} \right. \\ &\quad \left. - \left(\frac{e^{j\pi n}}{-j\pi n} - \frac{2e^{j\pi n}}{\pi^2 n^2} + \frac{2e^{j\pi n}}{j\pi^3 n^3} \right) \right] \\ &= \frac{1}{2} \left[\frac{4(-1)^n}{\pi^2 n^2} \right] = \frac{2(-1)^n}{\pi^2 n^2}, \end{aligned}$$

while

$$D_0 = \frac{1}{2} \int_{-1}^1 t^2 dt = \frac{1}{3}.$$

- (c) Represent $f(t)$ as a complex Fourier series.

Solution:

$$f(t) = \frac{1}{3} + \sum_{n \neq 0} \frac{2(-1)^n}{\pi^2 n^2} e^{jn\pi t}.$$

- (d) For the system with transfer function $H(s) = \frac{s}{s+2}$, find the system response to periodic input $f(t)$.

Solution:

$$H(jn\pi) = \frac{jn\pi}{jn\pi + 2}.$$

We can substitute the Fourier coefficients of $f(t)$ into the system response formula

$$y(t) = \sum_{n=-\infty}^{\infty} H(jn\pi) D_n e^{jn\pi t}.$$

Noting that when $n = 0$, $H(0) = 0$, causing the DC term to vanish, we can write

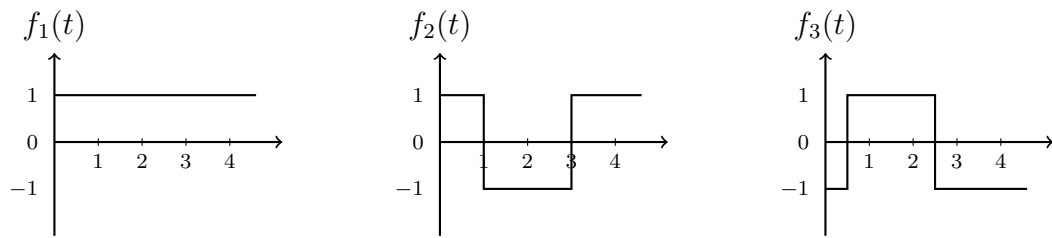
$$y(t) = \sum_{n \neq 0} \frac{jn\pi}{jn\pi + 2} \frac{2(-1)^n}{\pi^2 n^2} e^{jn\pi t}.$$

We can simplify this

$$y(t) = \sum_{n \neq 0} \frac{2j(-1)^n}{\pi n(2 + jn\pi)} e^{jn\pi t}.$$

6. Answer each of the following questions about orthogonality.

(a) True or false: the following functions $f_1(t)$, $f_2(t)$, and $f_3(t)$ are mutually orthogonal.



Solution: False.

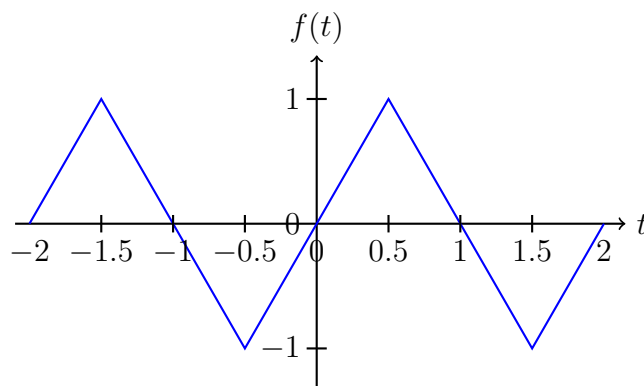
(b) True or false: the polynomials $f_1(t) = t$, $f_2(t) = t^2$, and $f_3(t) = t^3$ are mutually orthogonal on $t \in [-1, 1]$.

Solution: False.

(c) True or false: the functions $f_1(t) = t \cos(t)$ and $f_2(t) = t \sin(t)$ are orthogonal on $t \in [0, 2\pi]$.

Solution: False.

7. Find the trigonometric Fourier series of the function in the figure.



Please reduce your expression to its simplest form.

- (a) Find the fundamental frequency ω_0 .

Solution: The period is $T_0 = 2$, so

$$\omega_0 = \frac{2\pi}{T_0} = \pi.$$

- (b) Find the value a_0 .

Solution: The signal has odd symmetry, so $a_0 = 0$.

- (c) Find an expression for a_n .

Solution: The signal has odd symmetry, so $a_n = 0$.

- (d) Find an expression for b_n .

Solution: Using the positive half of the waveform,

$$\begin{aligned}
 b_n &= \frac{8}{2} \int_0^{0.5} 2t \sin(n\pi t) dt \\
 &= 8 \int_0^{0.5} t \sin(n\pi t) dt \\
 &= 8 \left[-\frac{t \cos(n\pi t)}{n\pi} + \frac{\sin(n\pi t)}{n^2\pi^2} \right]_0^{0.5} \\
 &= 8 \left[-\frac{0.5 \cos(n\pi/2)}{n\pi} + \frac{\sin(n\pi/2)}{n^2\pi^2} - 0 \right] \\
 &= 8 \left[\frac{\sin(n\pi/2)}{n^2\pi^2} - \frac{\cos(n\pi/2)}{2n\pi} \right].
 \end{aligned}$$

The cosine term is zero for odd n , so

$$b_n = \frac{8}{n^2\pi^2} \sin\left(\frac{n\pi}{2}\right).$$

(e) Represent $f(t)$ as a trigonometric Fourier series.

Solution:

$$f(t) = \sum_{n=1}^{\infty} \frac{8}{n^2\pi^2} \sin\left(\frac{n\pi}{2}\right) \sin(n\pi t).$$